

IT PROFESSIONAL PORTFOLIO

Windows Infrastructure and Administration Projects

PROJECT 01

Hyper-V Enterprise Lab Foundation

Hyper-V | Virtualization | Windows Server | Network Segmentation

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ENVIRONMENT	Personal Windows Enterprise Lab
VERSION	v1.0.0
RELEASE DATE	August 4, 2026
STATUS	Complete

Executive Summary

Built the reusable virtualization foundation for a Windows enterprise administration portfolio. The project enabled hardware virtualization and Hyper-V on a Windows 11 Pro workstation, established an organized lab storage structure, created isolated corporate and branch networks, assigned private host-side addresses, and provisioned three Generation 2 virtual machines with Windows Server 2025 and Windows 11 Pro.

Business Problem

Enterprise Windows labs require a controlled, repeatable environment before Active Directory, DNS, DHCP, Group Policy, file services, event collection, and forest trusts can be implemented safely. Building the hypervisor, storage layout, isolated networks, and base operating systems first separates platform setup from later identity and service configuration and creates a stable baseline for troubleshooting.

Objectives

- Enable processor virtualization in UEFI and install Microsoft Hyper-V.
- Create a consistent storage hierarchy for ISO media, VM configuration, VHDX files, and exports.
- Create isolated corporate and branch internal virtual switches.
- Assign private 172.16.0.0/12 addresses to the host-side vEthernet adapters.
- Provision secure Generation 2 VM shells for a domain controller, member server, and management client.
- Install and verify Windows Server 2025 Standard Evaluation and Windows 11 Pro without configuring domain services.

Environment

Component	Configuration
Host	HP Z2 Tower G9 Workstation Desktop PC
Host operating system	Microsoft Windows 11 Pro, 64-bit, build 26200
Processor	Intel Core i7-14700K - 20 cores / 28 logical processors
Installed memory	63.71 GB
Storage at assessment	475.86 GB total; 351.1 GB free on C:
Hypervisor	Microsoft Hyper-V with firmware virtualization enabled
Lab root	C:\HyperV-Lab
Network model	Two isolated Internal virtual switches
Project scope boundary	No AD DS, DNS, DHCP, domain join, or forest configuration

Architecture

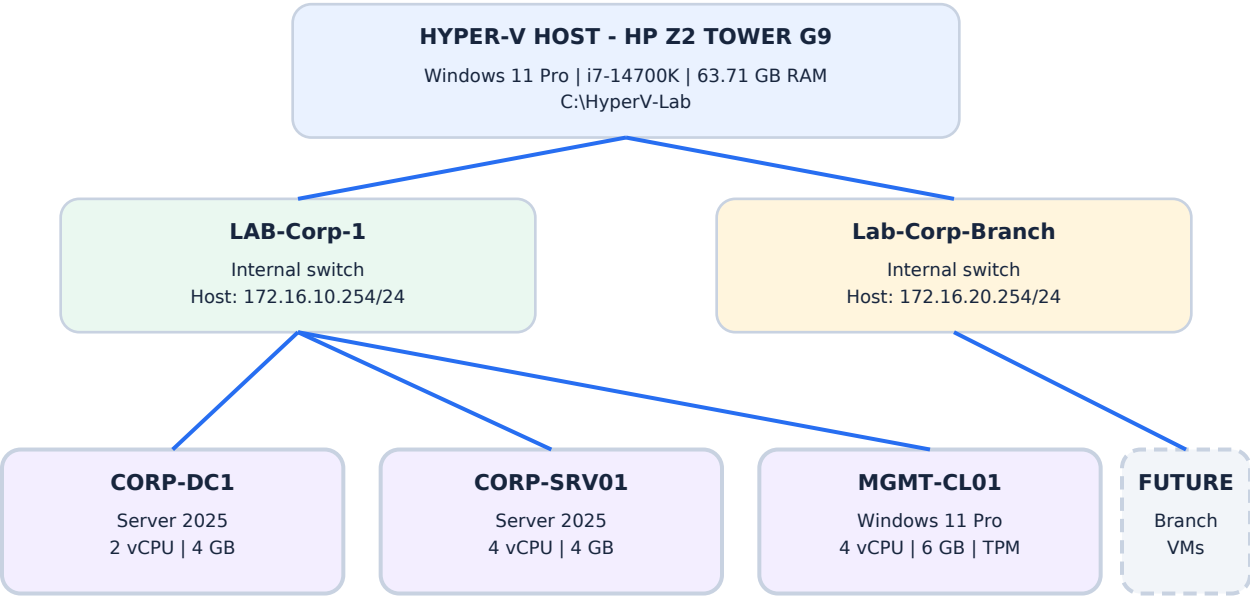


Figure 1 - Hyper-V host, isolated virtual switches, and initial corporate VM foundation.

Addressing Plan

Network / System	Address	Purpose
LAB-Corp-1	172.16.10.0/24	Corporate lab segment
Host vEthernet (LAB-Corp-1)	172.16.10.254/24	Host access to corporate lab
Lab-Corp-Branch	172.16.20.0/24	Reserved branch segment
Host vEthernet (Lab-Corp-Branch)	172.16.20.254/24	Host access to branch lab
Default gateway / DNS	Not assigned	Isolation maintained in Lab 01

Virtual Machine Inventory

VM	Guest OS	Compute	Security / Network
CORP-DC1	Windows Server 2025 Standard Evaluation	2 vCPU, 4096 MB	Secure Boot; LAB-Corp-1
CORP-SRV01	Windows Server 2025 Standard Evaluation	4 vCPU, 4096 MB	Secure Boot; LAB-Corp-1
MGMT-CL01	Windows 11 Pro	4 vCPU, 6144 MB	Secure Boot + TPM; LAB-Corp-1

Implementation Summary

- 1. Enabled Intel processor virtualization in the HP workstation UEFI firmware.
- 2. Installed and verified the Hyper-V Windows feature on Windows 11 Pro.
- 3. Created C:\HyperV-Lab with Exports, ISOs, VHDX, and VMs subfolders.
- 4. Configured Hyper-V to use the lab folder structure for VM and virtual disk storage.
- 5. Created LAB-Corp-1 and Lab-Corp-Branch as Internal virtual switches.
- 6. Configured the corresponding management-OS vEthernet adapters as 172.16.10.254/24 and 172.16.20.254/24 with no gateway or DNS server.
- 7. Created CORP-DC1, CORP-SRV01, and MGMT-CL01 as Generation 2 virtual machines on LAB-Corp-1.
- 8. Enabled Secure Boot on all three VMs, enabled virtual TPM for MGMT-CL01, selected Production-only checkpoints, and configured graceful guest shutdown.
- 9. Installed Windows Server 2025 Standard Evaluation on CORP-DC1 and CORP-SRV01.
- 10. Installed Windows 11 Pro on MGMT-CL01 and verified all three guest computer names.

Configuration Standards

Control	Applied Configuration	Operational Value
Generation 2 VMs	UEFI-based VM generation	Supports Secure Boot and modern Windows guests
Secure Boot	Enabled using Microsoft Windows template	Reduces unauthorized boot-code risk
Virtual TPM	Enabled for MGMT-CL01	Meets Windows 11 platform-security expectations
Checkpoints	Production-only	Uses application-consistent checkpoint behavior when supported
Automatic start	None	Prevents an unplanned lab workload start after host boot
Automatic stop	Shut down guest operating system	Provides controlled guest shutdown when the host stops
Network isolation	Internal switches; no gateway or DNS	Prevents unintended access to the physical network during setup

SCOPE	Lab 01 intentionally stops before static guest addressing, AD DS, DNS, DHCP, domain joining, or routing. Those changes begin in Lab 02 and later projects.
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Evidence - Hyper-V and Storage Foundation

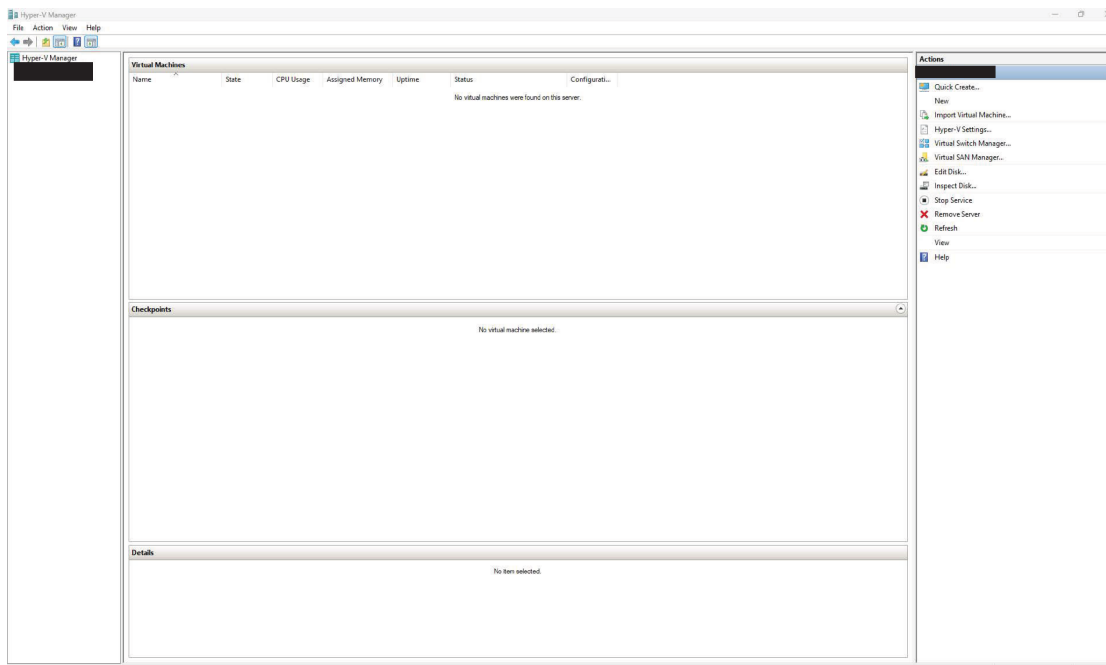


Figure 2 - Hyper-V Manager Installed and Operational

The redacted host capture shows Hyper-V Manager available and ready before VM provisioning.

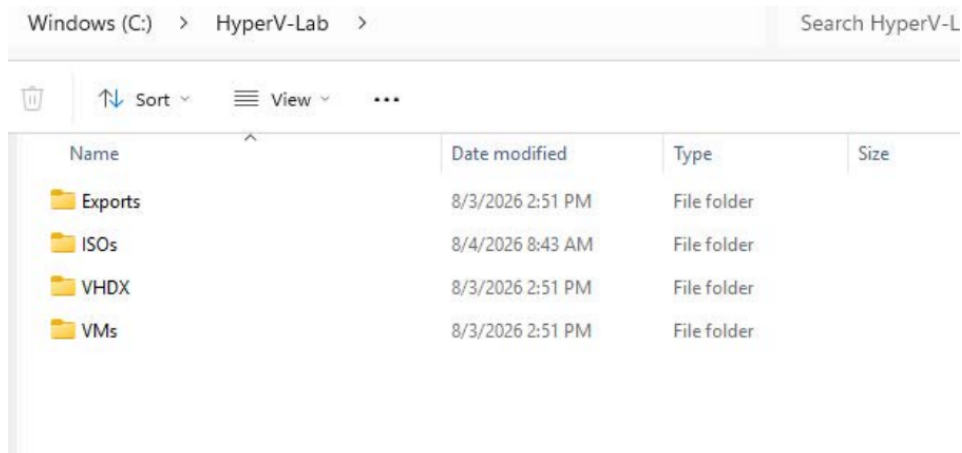


Figure 3 - C:\HyperV-Lab Storage Structure

File Explorer confirms the Exports, ISOs, VHDX, and VMs subfolders used to organize the lab.

Evidence - Isolated Virtual Networking

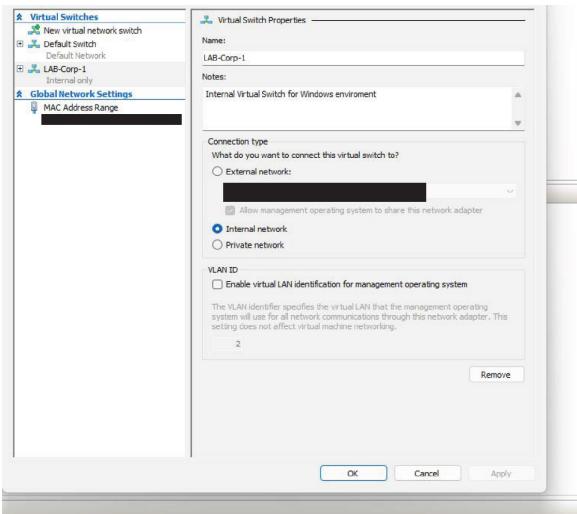


Figure 4 - Corporate Internal Switch LAB-Corp-1 is configured as Internal.

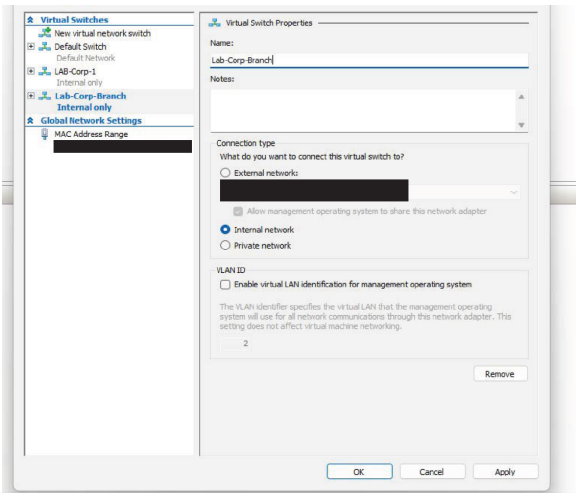


Figure 5 - Branch Internal Switch Lab-Corp-Branch is configured as Internal.

InterfaceAlias	IPAddress	PrefixLength
vEthernet (Lab-Corp-Branch)	172.16.20.254	24
vEthernet (LAB-Corp-1)	172.16.10.254	24

Figure 6 - Management-OS vEthernet Address Validation
PowerShell output confirms 172.16.10.254/24 for LAB-Corp-1 and 172.16.20.254/24 for Lab-Corp-Branch.

NETWORK

An Internal switch permits communication between connected VMs and the Hyper-V host. No external switch was used for these isolated lab segments.

Evidence - CORP-DC1 Base Server

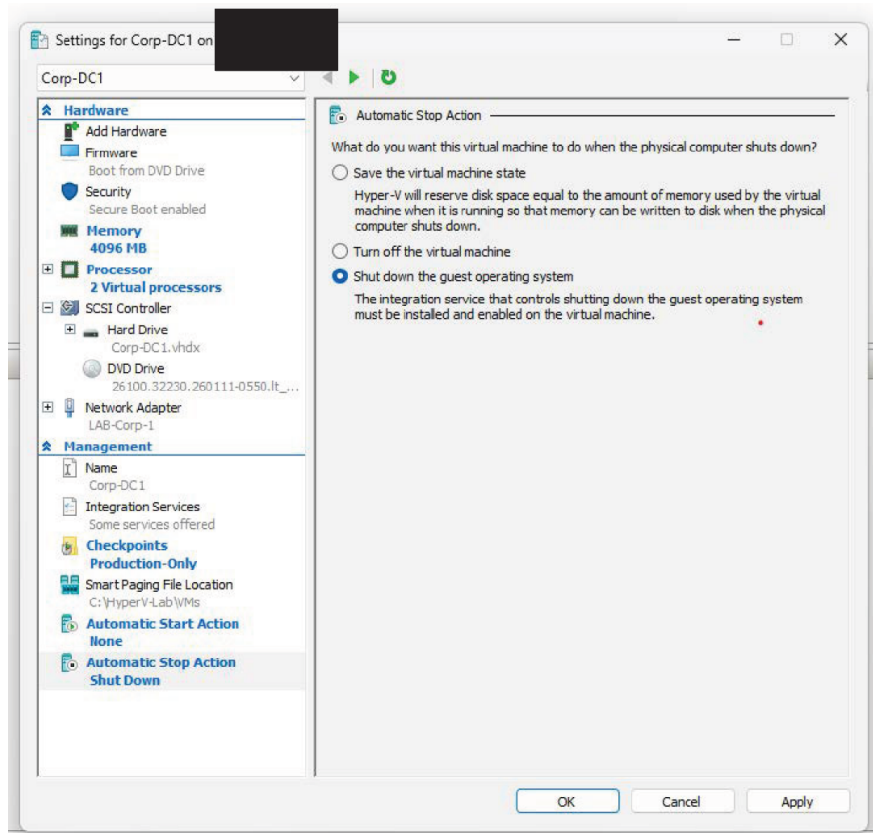


Figure 7 - CORP-DC1 Hyper-V Settings

The VM settings show Secure Boot, 4096 MB memory, 2 virtual processors, LAB-Corp-1 connectivity, Production-only checkpoints, no automatic start, and graceful guest shutdown.

```
Host Name: CORP-DC1
OS Name: Microsoft Windows Server 2025 Standard Evaluation
OS Version: 10.0.26100 N/A Build 26100
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Server
OS Build Type: Multiprocessor Free
```

Figure 8 - CORP-DC1 Operating-System Verification

System information confirms the CORP-DC1 name and Windows Server 2025 Standard Evaluation installation.

Evidence - CORP-SRV01 Base Server

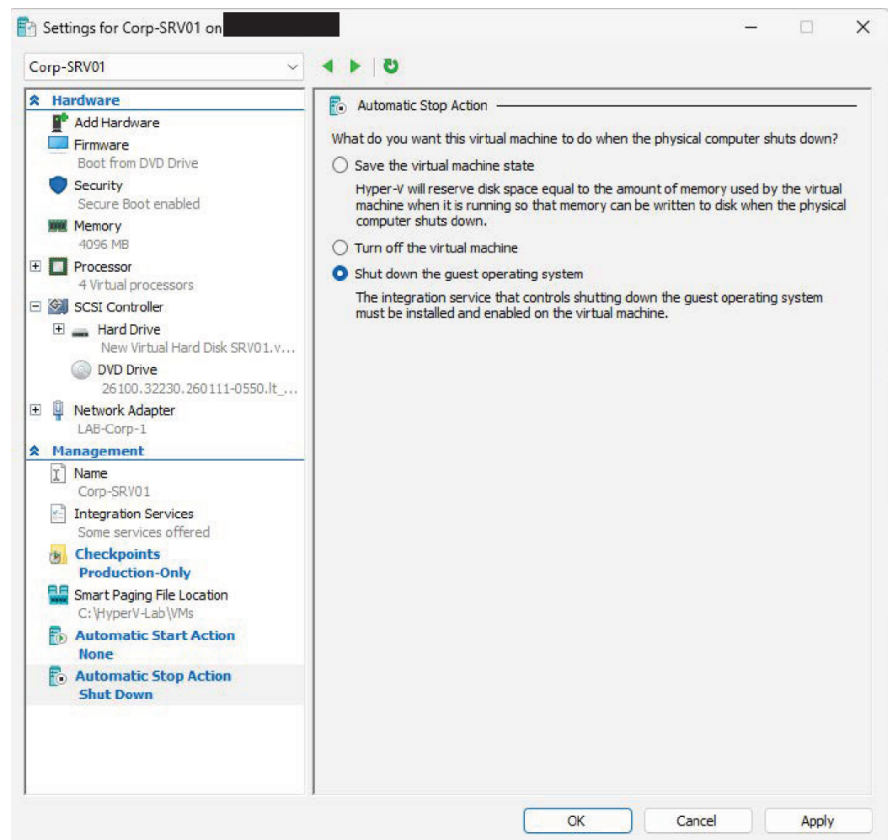


Figure 9 - CORP-SRV01 Hyper-V Settings

The VM settings show Secure Boot, 4096 MB memory, 4 virtual processors, LAB-Corp-1 connectivity, Production-only checkpoints, no automatic start, and graceful guest shutdown.

```
Host Name: CORP-SRV01
OS Name: Microsoft Windows Server 2025 Standard Evaluation
OS Version: 10.0.26100 N/A Build 26100
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Server
OS Build Type: Multiprocessor Free
```

Figure 10 - CORP-SRV01 Operating-System Verification

System information confirms the CORP-SRV01 name and Windows Server 2025 Standard Evaluation installation.

Evidence - MGMT-CL01 Management Client

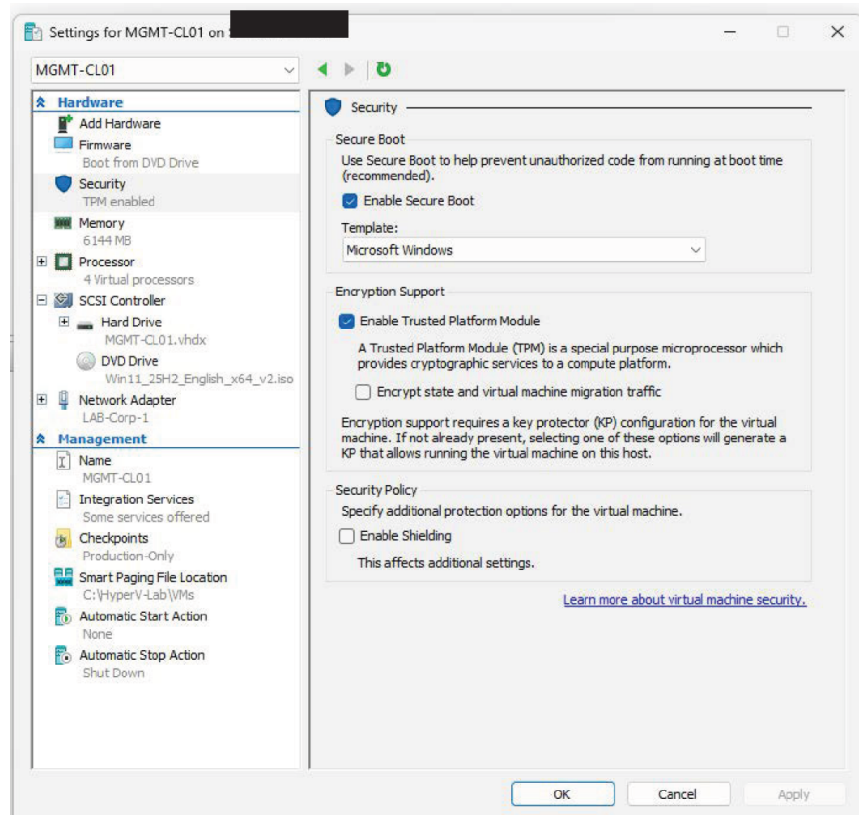


Figure 11 - MGMT-CL01 Hyper-V Security Settings

The settings confirm 6144 MB memory, 4 virtual processors, LAB-Corp-1 connectivity, Secure Boot, virtual TPM, Production-only checkpoints, and graceful guest shutdown.

```
Host Name: MGMT-CL01
OS Name: Microsoft Windows 11 Pro
OS Version: 10.0.26200 N/A Build 26200
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Workstation
OS Build Type: Multiprocessor Free
```

Figure 12 - MGMT-CL01 Operating-System Verification

System information confirms the MGMT-CL01 name and Windows 11 Pro installation.

Validation Results

Validation	Result	Evidence
Hyper-V management console available	PASS	Figure 2
Lab storage folders created	PASS	Figure 3
Corporate and branch switches are Internal	PASS	Figures 4-5
Host-side private addresses use the planned /24 subnets	PASS	Figure 6
CORP-DC1 baseline and Server 2025 installation	PASS	Figures 7-8
CORP-SRV01 baseline and Server 2025 installation	PASS	Figures 9-10
MGMT-CL01 Secure Boot, TPM, and Windows 11 Pro	PASS	Figures 11-12
AD DS, DNS, DHCP, routing, and domain configuration absent	PASS	Lab 01 scope boundary

Security Analysis

Using Internal switches limits the initial attack surface because the base VMs are not bridged directly to the physical LAN. This is especially valuable while operating systems are being installed and before Windows Firewall, patching, domain policies, and administrative access controls are configured.

Generation 2 VMs provide a modern UEFI-based platform. Secure Boot is enabled for every guest, and virtual TPM is enabled for the Windows 11 management client. Production-only checkpoints and graceful shutdown settings favor recoverability and consistent VM state over abrupt power-off behavior.

The separate corporate and branch subnets establish the segmentation needed for later routing, DNS forwarding, second-forest, and trust exercises without redesigning the base platform.

Evidence Limitations

The evidence set documents Hyper-V availability, storage paths, both switches, host vEthernet addressing, VM configuration, and installed guest operating systems. A separate screenshot of the UEFI virtualization setting and a post-build Hyper-V Manager inventory view were not included in this edition. The completed configuration is recorded from the implementation log and the supporting VM-specific evidence.

Lessons Learned

- Firmware virtualization and the Windows Hyper-V feature are separate prerequisites; both must be enabled before the hypervisor can run workloads.
- Hyper-V switch type determines reachability. Internal switches create host-to-VM and VM-to-VM connectivity without using the physical network.
- IP addresses are assigned to the management operating system's vEthernet adapters, not directly to the virtual switch object.
- Creating the folder, network, and VM standards before Active Directory reduces configuration drift across later labs.
- Guest names and visible settings should be verified before installing identity services because renaming a domain controller later creates unnecessary risk.

Production Improvements

- Store production VM files on redundant storage instead of the host operating-system volume.
- Apply a documented patching process to the Hyper-V host and every guest before enabling wider network access.
- Use dedicated management, storage, migration, and workload networks in a production virtualization cluster.
- Configure monitoring, backup, recovery testing, and capacity alerts for host CPU, memory, and storage.
- Use a naming standard and infrastructure automation to reproduce VM, switch, memory, processor, and disk configurations.
- Restrict membership in the local Hyper-V Administrators group and use separate privileged administration accounts.

Skills Demonstrated

Virtualization	Windows Infrastructure	Security / Operations
Hyper-V installation	Windows Server 2025 deployment	UEFI virtualization
Generation 2 VM provisioning	Windows 11 Pro deployment	Secure Boot and virtual TPM
Internal virtual switching	Private IPv4 subnet design	Isolated lab segmentation
VHDX and VM storage organization	System-information validation	Production checkpoints and graceful shutdown

Resume Bullet

RESUME	Designed and deployed a Hyper-V Windows enterprise lab on Windows 11 Pro, creating isolated corporate and branch networks, structured VM storage, and secure Generation 2 Windows Server 2025 and Windows 11 virtual machines with Secure Boot, TPM, production checkpoints, and validated private addressing.
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STAR Interview Story

Situation: I needed a safe, repeatable platform for demonstrating enterprise Windows administration without exposing unfinished servers to my physical home network.

Task: Build a virtualization foundation that could support Active Directory, DNS, DHCP, file services, Group Policy, centralized event collection, branch networking, and a future forest trust.

Action: I enabled firmware virtualization, installed Hyper-V, organized ISO and VM storage, created separate internal corporate and branch switches, assigned private host-side addresses, and deployed three secure Generation 2 VMs running Windows Server 2025 and Windows 11 Pro.

Result: The lab now has a documented and validated infrastructure baseline ready for the next project, while identity services and wider connectivity remain intentionally out of scope until their controls can be configured and tested.

Next Lab Handoff

Lab 02 begins with CORP-DC1: assign 172.16.10.10/24, install AD DS and DNS, create the corp.mslab.test forest, validate domain services, and then join the management client and member server in controlled stages.

Version History

Version	Release Date	Description
v1.0.0	August 4, 2026	Initial documented implementation. Enabled Hyper-V and UEFI virtualization, created the lab storage hierarchy, created two Internal switches and private host addresses, provisioned three secure Generation 2 VMs, installed Windows Server 2025 and Windows 11 Pro, and embedded the provided evidence using portfolio documentation standard v1.0.1.

References

- [Microsoft Learn - Install Hyper-V in Windows and Windows Server](#)
- [Microsoft Learn - Hyper-V host hardware requirements](#)
- [Microsoft Learn - Create and configure a virtual switch with Hyper-V](#)
- [Microsoft Learn - Create a virtual machine in Hyper-V](#)
- User-provided redacted Hyper-V, folder, virtual switch, vEthernet, VM settings, and operating-system evidence.